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Ejercito Mangawa Balay-odao

Department of Medicine, School of Medicine, Nazarbayev University, Kerey and Zhanibek Khans St 5/1, Astana, Kazakhstan, 010000 School of Advanced Studies, Saint Louis University, Baguio, Philippines, ejercito.balayodao@nu.edu.kz

Joseph Almazan

Department of Medicine, School of Medicine, Nazarbayev University, Kerey and Zhanibek Khans St 5/1, Astana, Kazakhstan, 010000

Jennifer Mesde

Department of Nursing Administration and Education, College of Nursing, Shaqra University, Dawadmi, Saudi Arabia

Junel Bryan Bajet

Department of Maternal and Child Nursing, College of Nursing, Shaqra University, Dawadmi, Saudi Arabia

Nahed Alquwez

Department of Nursing Administration and Education, College of Nursing, Shaqra University, Dawadmi, Saudi Arabia

See next page for additional authors

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ORIGINAL RESEARCH REPORTS

Influence of Clinical Supervision and Ethical Sensitivity on the Compassion Competence of Nursing Students

Ejercito M. Balay-odao ^{a,*} , Joseph Almazan ^a , Jennifer Mesde ^b , Junel B. Bajet ^c , Nahed Alquwez ^b , Cherryl C. Danglipen ^d , Jonas P. Cruz ^a 

^a Department of Medicine, School of Medicine, Nazarbayev University, Kerey and Zhanibek Khans St 5/1, Astana, 010000, Kazakhstan

^b Department of Nursing Administration and Education, College of Nursing, Shaqra University, Dawadmi, Saudi Arabia

^c Department of Maternal and Child Nursing, College of Nursing, Shaqra University, Dawadmi, Saudi Arabia

^d School of Nursing, Allied Health, and Biological Sciences, Saint Louis University, Baguio, Philippines

Abstract

Introduction: Clinical supervision is an essential component of nursing education. Ethical sensitivity and compassion competence are critical components to develop among nursing students to ensure patient-centered nursing care.

Aim: This study aimed to determine clinical supervision, ethical sensitivity, and compassion competence among student nurses and its influencing factors. Also, it determined the influence of clinical supervision and ethical sensitivity on compassion competence.

Method: This study utilized a descriptive quantitative cross-sectional correlation design. Data were collected from 452 nursing students from August 2022 to January 2023 using convenient sampling. The “Clinical Supervision Evaluation Questionnaire,” the “Ethical Sensitivity Questionnaire for Nursing Students,” and the “Compassion Competence Scale Arabic version” were used. The study employed multiple regression analysis to examine the influence of clinical supervision and ethical sensitivity on compassion competence.

Results: The students' overall mean on “Clinical supervision,” “Ethical sensitivity,” and “Compassion competence” was 4.06 ($SD = 0.77$), 3.66 ($SD = 0.50$), and 4.21 ($SD = 0.72$), respectively. The analysis revealed a weak and positive correlation between the students' age and clinical supervision perception ($r = 0.20$, $p < 0.001$). A unit increase in the mean scores of “Process” and “Impact” was associated with 0.17 ($p = 0.001$, 95 % CI = 0.07, 0.26) and 0.20 ($p < 0.001$, 95 % CI = 0.14, 0.27) unit increase in the “Ethical sensitivity” mean scores. Female students had higher levels of compassion competence than men ($\beta = 0.15$, $p < 0.001$, 95 % CI = 0.08, 0.22). A unit increase in the mean scores in “Process,” “Impact,” and “Ethical sensitivity” were linked to 0.42 ($p < 0.001$, 95 % CI = 0.33, 0.51), 0.12 ($p < 0.001$, 95 % CI = 0.06, 0.18), and 0.38 ($p < 0.001$, 95 % CI = 0.30, 0.47) unit increase in the compassion competence mean score.

Conclusion: Student nurses' perceptions of clinical supervision, ethical sensitivity, and compassionate competence vary. Factors that affect the student nurses' perception of ethical sensitivity are gender, academic level, and clinical supervision. The connection between compassion competence among nursing students and the ethical sensitivity of the healthcare profession is inherent. Nursing students who comprehend ethical considerations are more likely to demonstrate compassion.

Keywords: Clinical supervision, Compassion, Ethical sensitivity, Student nurses

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* Corresponding author.

E-mail addresses: ejercito.balayodao@nu.edu.kz (E.M. Balay-odao), joseph.almazan@nu.edu.kz (J. Almazan), jmesde@su.edu.sa (J. Mesde), bryanbajet@su.edu.sa (J.B. Bajet), nalquwez@su.edu.sa (N. Alquwez), ccdanglipen@slu.edu.ph (C.C. Danglipen), jonas.cruz@nu.edu.kz (J.P. Cruz).

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1. Introduction

In healthcare, particularly nursing, compassion competence is paramount for providing effective and empathetic patient care. However, the development of this crucial competency is influenced by various factors, including clinical supervision and ethical sensitivity. Clinical supervision provides a structured framework for guiding and supporting nursing students in their clinical practice, while ethical sensitivity enhances their ability to recognize and navigate ethically complex situations [1].

In Saudi Arabia, nursing education and practice are undergoing rapid development. Nursing students are required to have a thorough understanding of clinical supervision, ethical sensitivity, and compassion competency. This study determines compassion competency, clinical supervision, and ethical sensitivity in the context of nursing practice and education in Saudi Arabia. Determining these aspects will help Saudi student nurses enhance their ability to provide ethical sensitivity and compassionate care by utilizing the research findings to develop tailored learning strategies and additional support mechanisms. Studying nursing education in Saudi Arabia can offer important insights into the key characteristics of compassion competency, ultimately enhancing nursing students' clinical guidance and ethical accountability.

1.1. Background

Nursing education shapes future nursing professionals' competencies, attitudes, and perceptions [2,3]. In nursing education, clinical supervision [4,5], ethical sensitivity [6], and compassion competence [7,8] are essential in developing the student nurses' knowledge, skills, and attitudes. Clinical supervision is central to nursing education, which is also the cornerstone of nursing students to develop and practice effective and compassionate patient care [1]. Clinical supervision in nursing education is a structured process that facilitates students' theoretical integration of knowledge with practical skills in clinical settings [5]. It involves guidance, feedback, and support from experienced nurse educators or preceptors to enhance students' clinical competence and confidence [5]. It is recognized that sufficient clinical supervision offers students the opportunity to build reflective learning skills and apply evidence-based practice in real-life situations. It also fosters a supportive learning environment where students can seek clarification, discuss challenges, and receive constructive feedback to improve their clinical performance [5].

Ethical sensitivity is the cornerstone of effective and compassionate patient care for future nurses [1]. Weaver [6] defined ethical sensitivity as the initial step in assessing, gathering, and deciding on alternative solutions in professional practice. It involves the ability to identify and understand the ethical aspects of each care setting. Ethical sensitivity involves applying ethical principles, such as beneficence, nonmaleficence, autonomy, and justice, to navigate ethical dilemmas effectively [9]. Nursing students' perceptions of ethical sensitivity influence their ethical decision-making processes, professional conduct, and moral development as future healthcare professionals [10,11].

Compassion competence is also a fundamental aspect of nursing practice, reflecting the ability to empathize with patients, demonstrate kindness and understanding, and deliver comprehensive care that attends to patients' physical, emotional, and psychosocial needs [7,8]. Alquwez et al. [12] elucidated that nursing students' perceptions of compassion competence influence their attitudes, behaviors, and interactions, impacting the quality of care delivery and patient outcomes. A study conducted in Saudi Arabia found that the compassion competence of student nurses significantly influences their choice of being a nurse as their career [12]. Thus, academics, clinical instructors, and healthcare facilities should intentionally mold the compassion competence of student nurses in clinical settings.

Saudi Arabia has significantly revised its nursing curriculum in response to the growing need to comply with international nursing standards. The Saudi Commission for Health Specialties (SCFHS) regulates nursing education in Saudi Arabia and establishes standards, rules, and guidelines for nursing education programs. Furthermore, it meticulously assesses the qualifications of professors and instructors and provides comprehensive information about clinical training [13]. The SCFHS [13] affirms that nursing schools adhere to the benchmarks established by the international community and effectively prepare students to address Saudi Arabia's healthcare requirements.

Previous studies in the country have emphasized the essence of clinical supervision, ethical decision-making, and compassion competence in nursing education [14–16]. These components enhance patient-centered nursing care by facilitating the practical application of academic knowledge in healthcare. Moreover, the recently published “Scope of Nursing and Midwifery Practice” in Saudi Arabia laid down the “full spectrum of functions, responsibilities, roles, activities, and decision-making capacity that nurses and midwives are educated

for, competent in, and authorized to perform” [17]. This document clarifies the competencies nurses in the country should possess and emphasizes the different domains of the nurses’ competence. Domains 1 and 2 highlight “Professionalism and Ethical Practice” and “Patient-Centered Care,” respectively [17]. Ethical sensitivity and compassion competence are critical in achieving ethical practice and patient-centered care. Therefore, this new scope of nursing practice challenges nursing education in the kingdom to prepare future nurses who meet the standards of nursing practice within the identified scope of nursing practice.

The availability of knowledge on clinical supervision, ethical sensitivity, and compassion competency in Saudi nursing schools is limited. Nevertheless, nursing education improved throughout the kingdom when basic nursing education was moved to institutions with bachelor's degrees [18,19]. Nursing students' experiences in the clinical setting influence their development of compassion competencies, with positive aspects like patient acceptance and motivation for professional growth [20]. Clinical group supervision stimulates reflection and discussion on ethical situations, revealing underlying caring actions and improving nursing education and practice [21]. Clinical nursing supervision leads to developing ethical competence through reflection on ethically difficult caring situations, integrating nursing theory and practice [22]. Although these aspects are highly valued in nursing education, there is a dearth of empirical studies investigating the influence of clinical supervision on nursing students' ethical sensitivity and compassion competence.

Saudi Arabian nursing schools include ethics in their curriculum to enhance nursing students' knowledge and comprehension in handling real-life moral challenges effectively or discern the motivating factors that foster moral awareness [13]. Moral sensitivity and professional values positively correlate with ethical decision-making in nursing students, with professional values having a mediating effect [23]. Nursing students believe that ethical supervision is essential, as it helps them feel welcome, responsible, and treated as individuals, which can influence their compassion competence [24]. This study examines the correlation between compassion competency, clinical supervision, and ethical sensitivity in the context of nursing practice and education in Saudi Arabia. Prospective Saudi nurses could enhance their ability to provide compassionate care by utilizing the research findings to develop tailored learning strategies and additional support mechanisms. Examining nursing

education in Saudi Arabia can provide valuable insights into the defining attributes of compassion competency, thereby fostering nursing students' professional growth and ethical responsibility. Similarly, a correlation between compassion competency, clinical supervision, and ethical sensitivity among nursing students was conducted in Korea [25], China [26], and Turkey [27]. These studies highlight the importance of ethics and advocacy education in enhancing nursing students' competencies, with cultural contexts influencing the expression and development of these skills [25–27].

1.2. Objectives

The objectives of the study are to:

1. Assess student nurses' level of clinical supervision, ethical sensitivity, and compassion competence.
2. Examine the impact of student nurses' demographic data on clinical supervision, ethical sensitivity, and compassion competence.
3. Investigate the influence of clinical supervision on the ethical sensitivity of nursing students.
4. Explore the effect of clinical supervision and ethical sensitivity on compassion competence among student nurses.

1.3. Hypotheses

1. The students' demographic data influences clinical supervision, ethical sensitivity, and compassion competence.
2. Clinical supervision of student nurses influences the ethical sensitivity practice.
3. Clinical supervision and ethical sensitivity influence the compassion competence of student nurses.

2. Method

2.1. Study approach and design

This study used a descriptive quantitative cross-sectional correlation design and complied with STROBE guidelines (Supplemental file 1 (https://hpe.researchcommons.org/cgi/editor.cgi?article=1352&window=additional_files&context=journal)) [28].

2.2. Setting

The survey was conducted in Riyadh Region, Saudi Arabia Government School. The university has three campuses in three different cities in the

Riyadh region: one in the northwest, the second in the south, and the third in the southwest of the Riyadh region. Each campus offers various academic programs, including nursing, engineering, and business, with separate male and female campuses. The university is part of Saudi Arabia's educational reforms, aiming to enhance higher education and meet the needs of sectors like healthcare and technology. The Bachelor of Science in Nursing (BSN) program at this University combines theoretical and practical courses to prepare graduates for successful careers in healthcare. The Nursing program at this university offers a robust nursing curriculum that combines theoretical and practical courses to provide high-quality nursing education and prepare nurse graduates for successful careers in the healthcare industry.

2.3. Sampling process

Convenient sampling was applied to include all three campuses, as they all offer nursing programs and have nursing students enrolled at various academic levels.

The study confirmed the inclusion of all three campuses in the sample, ensuring that data from each campus were represented proportionally. The total population of nursing students across the three campuses was 498, and a sample size of 452 students was selected using convenient sampling. The number of students who participated from each campus was based on the available population and accessibility: 84 nursing students from Campus A participated out of 97 population, 109 nursing students were selected from campus B out of 126, and 259 were selected from campus B out of 275 nursing students.

The data were collected from the respondents that meet the following inclusion criteria: (1) Saudi citizens; (2) nursing students from second to fourth year of study since they are being exposed in the clinical area; (3) individuals aged 18 and above; (4) students who are enrolled full-time in courses; and (5) those who have willingly expressed their interest in participating. Students who were on academic leave or had not participated in clinical rotations were excluded from the sample.

2.4. Instruments

Three instruments, “The Clinical Supervision Evaluation Questionnaire (CSEQ),” the “Ethical Sensitivity Questionnaire for Nursing Students (ESQ-NS),” and the “Compassion Competence Scale Arabic version (SSC-A),” were used after the

authors granted permission. The instruments utilized in this study are written in English. In addition, the demographic profiles (age, gender, year level, family structure, and grade point average) of the respondents were obtained.

2.4.1. Clinical supervision evaluation questionnaire

The CSEQ utilizes 14 statements that evaluate the student's perception of clinical supervision. The CSEQ has three subscales, namely, “Purpose,” “Process,” and “Impact.” The respondents were asked to rate their agreement on the 14 statements using the five-point Likert scale (strongly disagree to strongly agree). Mean scores were computed, with higher means indicating better perceptions of the clinical supervision they receive. Cronbach's alpha was used to calculate the internal consistency value of 0.86. The scale underwent a factor analysis, which supported the three-component structure, explaining 72 % of its variability [29]. In the study sample, the Cronbach's alpha of the entire scale was 0.944, while the Cronbach's alpha for the three subscales ranged from 0.867 to 0.910.

2.4.2. Ethical sensitivity questionnaire for nursing students

The ESQ-NS survey is designed to independently investigate moral difficulties nursing students encounter throughout their clinical practice within the ethical awareness framework. The questionnaire utilized a four-point Likert scale consisting of 13 questions. The instruments consist of three subscales: “respect for individuals with eight items, distributive justice with three items, and maintaining patient confidentiality with two items.” Mean scores were calculated with high means indicating high ethical sensitivity. The tool reliability was 0.82 using Cronbach's alpha [30]. In the present sample, the ESQ-NS's Cronbach's alpha was 0.924. The Cronbach's alpha ranged from 0.817 to 0.858 for the three subscales.

2.4.3. Compassion competence scale Arabic version

The CCS-A was used to assess the nursing student's perceived compassion competence. The tool has 17 questions with three subscales: “insight,” “sensitivity,” and “communication.” The perceived compassion competence of nursing students was measured using the five-point Likert scale, from “strongly disagree” to “strongly agree.” A high average score across all subscales indicates student nurses' significant perceived compassion competence. The reliability of the instrument using Cronbach's alpha coefficient was 0.806. The reliabilities of the subscales “insight,” “communication,” and

“sensitivity” were $\alpha = 0.797, 0.788, \text{ and } 0.739$, respectively. The stability and dependability using the “intraclass correlation coefficient” was 0.84 [16]. A Cronbach's alpha of 0.955 was calculated for the entire scale in this study. Cronbach's alpha, ranging from 0.896 to 0.945, was computed for the three subscales.

2.5. Data collection

The respondents were identified and approached based on inclusion criteria. A detailed explanation of the study's purpose, objectives, and procedures was provided to all respondents. Written informed consent was obtained. The questionnaires were distributed physically at the participating University. To minimize disruption, questionnaires were distributed after the respondents' class sessions. Respondents were provided with the principal investigator's contact information for any inquiries or concerns about the study, including questions about its methodology, risks, or benefits. Data collection occurred from August 2022 to January 2023. The researchers supervised the data collection process to ensure consistency and adherence to the protocol. Upon completing the data collection process, the responses were systematically tallied, tabulated, and analyzed in preparation for subsequent statistical evaluation.

2.6. Data analysis

The data were examined using descriptive statistical measures such as frequency, percentage, mean, and standard deviation. Statistical analyses included independent samples t-test, One-way Analysis of variance with Tukey HSD test to compare variables, and Pearson product-moment correlations to examine the association between variables. Two multiple linear regression analyses were conducted in the study to examine the influence of variables. P values below .05 were considered statistically significant. Multiple linear regression was used in the study because the target variable is continuous, the relationship between the variables is assumed to be linear, the model is interpretable and straightforward to implement, the assumptions of linear regression are met in the data, and it provides a balance of predictive power and simplicity. SPSS Version 24.0 was used for all the analyses.

2.7. Ethical consideration

The Institutional Review Board of Higher Education approved the study (ERC-20210041, March 6,

2022). The informed consent agreement covers the following aspects: study goals, voluntary participation, confidentiality, autonomy, and the right to withdraw. Their involvement or absence in this study will not affect their academic progression. However, the investigation was conducted to improve understanding of ethical sensitivity, clinical supervision, and compassion competency. The outcomes of their research will not influence their performance assessment. Respondents were prompted to disclose any unfavorable encounters while collecting data. Survey respondents were provided a succinct overview of the data collected from this investigation. Collected data were anonymized to maintain participant confidentiality. Data were securely stored in a password-protected system accessible only to the research team.

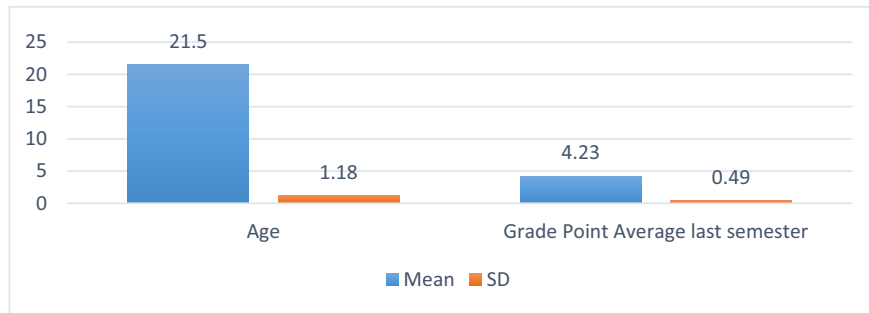
3. Results

3.1. Demographic characteristics of the respondents

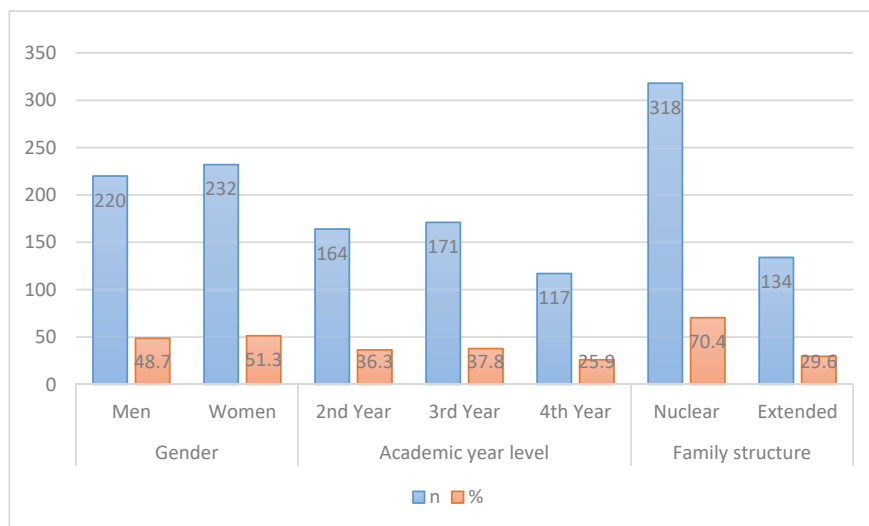
The mean age of the respondents was 21.50 years ($SD = 1.81$). Most respondents were women ($n = 232$) and from a nuclear family ($n = 318$). There were 37.8 % ($n = 17$) respondents from the 3rd year of the four-year BSN program, while 36.3 % ($n = 164$) and 25.9 % ($n = 117$) of the students were in the 2nd and 4th years, respectively. The mean of the student's grade point average was 4.23 ($SD = 0.49$) (Graph 1, 2).

3.2. Descriptive result of clinical supervision, ethical sensitivity, and compassion competence of nursing students

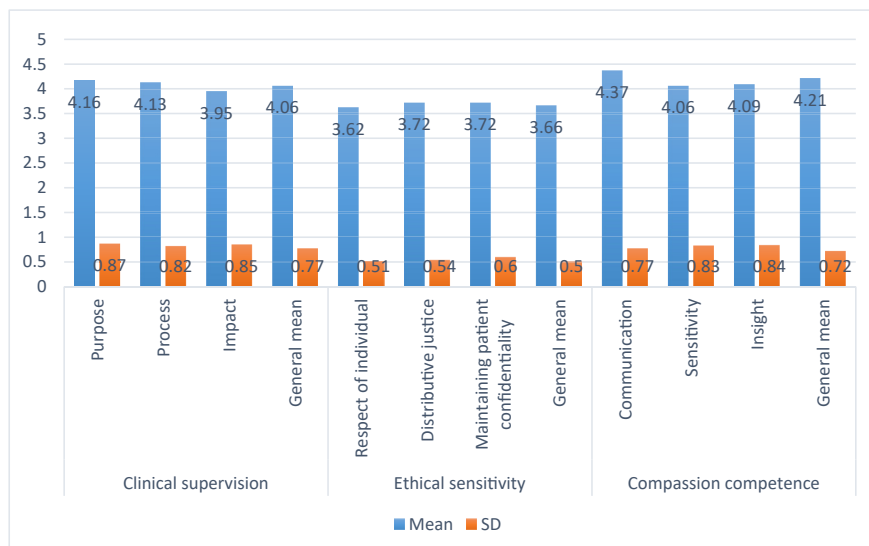
The students' overall mean on “Clinical supervision,” “Ethical sensitivity,” and “Compassion competence” was 4.06 ($SD = 0.77$), 3.66 ($SD = 0.50$), and 4.21 ($SD = 0.72$), respectively. The students rated “Purpose” as the highest subscale of “Clinical supervision” with a mean score of 4.16, ($SD = 0.87$), followed by “Process” ($mean = 4.13, SD = 0.82$). The lowest mean score was recorded in “Impact” ($mean = 3.95, SD = 0.85$). For “Ethical sensitivity,” “Distributive justice” ($mean = 3.72, SD = 0.54$) and “Maintaining patient confidentiality” ($mean = 3.72, SD = 0.60$) received the highest mean score, while “Respect of individual” ($mean = 3.62, SD = 0.51$) received the lowest mean score. The students reported high competence in the three dimensions of compassion competence, with “Communication” receiving the highest mean score of 4.37 ($SD = 0.77$), followed by “Insight” ($mean = 4.09, SD = 0.84$) and “Sensitivity” ($mean = 4.06, SD = 0.83$) (Graph 3).



Graph 1. Age and grade point average last semester of the respondents ($n = 452$).



Graph 2. Demographic characteristics of the respondents ($n = 452$).



Graph 3. Descriptive statistics results on the study variables ($n = 452$).

3.3. Demographic factors associated with the students' clinical supervision

The analysis revealed a weak and positive correlation between the students' age and clinical supervision perception ($r = 0.20$, $p < 0.001$). There was a statistical difference in the clinical perceptions of students between year levels ($F = 5.83$, $p = 0.033$), with 4th-year students ($mean = 4.27$, $SD = 0.73$) reporting better perceptions than 2nd ($mean = 4.00$, $SD = 0.74$, $p = 0.011$) and 3rd-year students ($mean = 3.98$, $SD = 0.81$, $p = 0.005$). A weak and positive relationship also existed between the students' grade point average (GPA) and their clinical supervision perceptions ($r = 0.10$, $p = 0.036$) (Table 1).

Table 1. Demographic factors associated with the students' clinical supervision ($n = 452$).

Variable	Mean	SD	Statistical test	p
Age			$r = 0.20$	$<0.001^{***}$
Gender				
Men	4.13	0.85	$t = 1.82$	0.069
Women	3.99	0.69		
Academic year level ^a			$F = 5.83$	0.003**
2nd year	4.00	0.74		
3rd year	3.98	0.81		
4th year	4.27	0.73		
Family structure				
Nuclear	4.10	0.78	$t = 1.76$	0.078
Extended	3.96	0.76		
Grade Point Average last semester			$r = 0.10$	0.036*

*Significant at 0.05 level, **Significant at 0.01 level, ***Significant at 0.001 level.

^a 4th year >2nd year ($p = .011$) and 3rd year ($p = .005$).

3.4. Influence of clinical supervision on the ethical sensitivity of nursing students

The multiple linear regression model with the ethical sensitivity as the dependent variable was statistically significant ($F_{9, 442} = 36.81$, $p < 0.001$), explaining 41.7 % ($R^2 = 0.428$, Adjusted $R^2 = 0.417$) of the variance in the students' ethical sensitivity. As shown in Table 2, gender, academic year level, and clinical supervision significantly influenced the students' ethical sensitivity. Female students exhibited better ethical sensitivity than men ($\beta = 0.16$, $p < 0.001$, 95 % CI = 0.09, 0.24). Second-year students had poorer ethical sensitivity than fourth-year nursing students ($\beta = -0.12$, $p = 0.028$, 95 % CI = -0.22 , -0.01). A unit increase in the mean scores of "Process" and "Impact" was associated with 0.17 ($p = 0.001$, 95 % CI = 0.07, 0.26) and 0.20 ($p < 0.001$, 95 % CI = 0.14, 0.27) unit increase in the "Ethical sensitivity" mean scores. This finding indicates a positive influence of effective clinical supervision on students' ethical sensitivity.

3.5. Influence of clinical supervision and ethical sensitivity on compassion competence of nursing students

The regression model of the influence of clinical supervision and ethical sensitivity on the students' compassion competence is reflected in Table 3. The model was statistically significant ($F_{10, 441} = 140.49$, $p < 0.001$), explaining 75.6 % ($R^2 = 0.761$, Adjusted $R^2 = 0.756$) of the overall variance in the students' compassion competence. According to the model, gender, "process," "impact," and "ethical sensitivity" were significant factors influencing the students' compassion competence. Female students had higher levels of compassion competence than men

Table 2. Influence of clinical supervision on ethical sensitivity of nursing students ($n = 452$).

Predictor variables	β	SE-b	Beta	t	p	95 % CI	
						Lower	Upper
Gender	0.16	0.04	0.16	4.22	$<0.001^{***}$	0.09	0.24
Age	-0.02	0.01	-0.08	-1.86	0.064	-0.04	0.00
Academic year level (Ref.: 4 th year)							
2nd year	-0.12	0.05	-0.11	-2.20	0.028*	-0.22	-0.01
3rd year	-0.01	0.05	-0.01	-0.20	0.841	-0.11	0.09
Family structure	0.03	0.04	0.03	0.86	0.389	-0.04	0.11
GPA last semester	0.04	0.04	0.04	1.04	0.298	-0.04	0.12
Clinical supervision							
Purpose	0.04	0.04	0.08	1.02	0.308	-0.04	0.13
Process	0.17	0.05	0.28	3.51	0.001**	0.07	0.26
Impact	0.20	0.03	0.34	6.25	$<0.001^{***}$	0.14	0.27

Note. The dependent variable was ethical sensitivity. B is the unstandardized coefficient; SE-b is the Standard Error.

$R^2 = 0.428$, Adjusted $R^2 = 0.417$.

*Significant at 0.05 level, **Significant at 0.01 level, ***Significant at 0.001 level.

Table 3. Influence of clinical supervision and ethical sensitivity on compassion competence of nursing students ($n = 452$).

Predictor variables	β	SE-b	Beta	t	p	95 % CI	
						Lower	Upper
Gender	0.15	0.04	0.10	4.11	<0.001***	0.08	0.22
Age	0.00	0.01	0.01	0.43	0.669	−0.02	0.03
Academic year level (Ref.: 4 th year)							
2nd year	0.00	0.05	0.00	0.03	0.977	−0.10	0.10
3rd year	0.04	0.05	0.03	0.95	0.345	−0.05	0.13
Family structure	−0.06	0.04	−0.04	−1.49	0.137	−0.13	0.02
GPA last semester	−0.02	0.04	−0.01	−0.55	0.582	−0.09	0.05
Clinical supervision							
Purpose	0.08	0.04	0.09	1.88	0.061	0.00	0.15
Process	0.42	0.04	0.48	9.36	<0.001***	0.33	0.51
Impact	0.12	0.03	0.14	3.83	<0.001***	0.06	0.18
Ethical sensitivity	0.38	0.04	0.27	8.68	<0.001***	0.30	0.47

Note. The dependent variable was compassion competence. B is the unstandardized coefficient; SE-b is the Standard Error.

$R^2 = 0.761$, Adjusted $R^2 = 0.756$.

***Significant at 0.01 level.

($\beta = 0.15$, $p < 0.001$, 95 % CI = 0.08, 0.22). A unit increase in the mean scores in “Process,” “Impact,” “and Ethical sensitivity” were linked to 0.42 ($p < 0.001$, 95 % CI = 0.33, 0.51), 0.12 ($p < 0.001$, 95 % CI = 0.06, 0.18), and 0.38 ($p < 0.001$, 95 % CI = 0.30, 0.47) unit increase in the compassion competence mean score.

4. Discussion

This study examined clinical supervision, ethical sensitivity, and compassion competence and determined the factors that influence them. It also examined the influence of clinical supervision and ethical sensitivity on the compassion competence of student nurses. The essential results of the study on clinical supervision, ethical sensitivity, compassion competence, and the factors that influence them are discussed.

The first essential study results show that nursing students highly rated clinical supervision. The nursing school's supervision program helps nursing students develop and integrate knowledge by enhancing their professional, interpersonal, and communication skills, which may explain the high mean clinical supervision score [22]. According to Pitkanen et al., [31] supportive clinical supervision and individualized, unbiased supervision considerably enhance nursing students' development and expertise.

The current study indicated that fourth-year and older nursing students significantly influence the clinical supervision perspective. Qumer and Fatima [32] and Şahin et al. [33] found that nursing students' age and year of study influenced their opinions of clinical learning settings. Senior students may find expressing their needs easier and seeking

help under clinical supervision. They can also feel more comfortable expressing their opinions to superiors [5].

Furthermore, older nursing students may perceive higher clinical supervision than younger nursing students. This finding could be associated with the various clinical preparation and courses that significantly influence their comprehension of the importance of clinical supervision in their clinical responsibilities [34]. Further, older nursing students are satisfied with facilitators' supervision, with preceptors perceived as more supportive and clinical lecturers as more important for challenging critical thinking and reflection [35]. Older nursing students value supervision, a pedagogical environment, and individualized supervision, with frequent bedside rounds important [36].

Another study found a weak and positive relationship between the nursing students' latest GPA and their clinical supervision perceptions. This finding could be associated with the notion that students with high GPAs value the guidance of both preceptors and instructors during clinical practice; they place greater emphasis on the impact of instructor supervision on their learning outcomes [37]. Similarly, Dube and Mlotshwa [38] stated that good nurse-educator relationships, academic support services offered by teachers, and the availability of qualified nurse educators improve students' academic performance.

The second essential study result shows that nursing students rated ethical sensitivity high. Student nurses' high ethical sensitivity score could be attributed to their heightened moral awareness as they advance in their education. They prioritize honesty, patient safety, and respect as crucial values [39]. Moreover, this is the ethical framework of

nursing, as these values are integral to providing high-quality nursing care. Standardized patient care greatly enhances the ethical sensitivity of student nurses; however, ethical reasoning and decision-making are low [40].

The current study shows that the student nurses' ethical sensitivity is influenced by their academic year level, where 4th year nursing students have a better perception of ethical sensitivity than 2nd-year nursing students. Nursing students encounter ethical problems in clinical settings throughout their academic careers [41]. Fourth-year nursing students have greater clinical experience and are more aware of professional norms and nursing ethics [42]. Higher academic year levels may help identify and address ethical issues in patient care [43]. Higher academic year levels, self-confidence, and ethical sensitivity help nursing students handle ethical issues [44].

This study shows that nursing students' gender influences their ethical sensitivity. Gender is one of several social, cultural, and psychological aspects that influence ethical sensitivity. Kim [43] states that gender affects nursing students' ethical sensitivity. Also, it was found that females may react differently to ethical dilemmas due to gender roles [45]. Female nursing students had more distinctive perspectives, communication approaches, and problem-solving ways than males. Empathy and connection abilities are higher in female, which can increase their awareness of healthcare ethics [46]. Mosallanezhad et al. [47] states that female nursing students are more ethically sensitive, while male students are more emotionally intelligent. Female nursing students considered ethical values to be the most important, followed by professional expertise and devotion [48].

Another study finding shows that clinical supervision influences student nurses' ethical sensitivity. Clinical supervision is the most effective and efficient method for promoting ethical awareness among nursing students. Facilitating patients' ability to contemplate, engage in conversation, and seek guidance addressing moral predicaments during clinical practice is essential to adequate supervision [5]. The quality of clinical supervision could impact nursing students' ethical growth. Supervisors with expertise, understanding, and support for ethical concepts can help students develop their ethical awareness by exemplifying ethical behavior, offering constructive feedback, and encouraging ethical decision-making [49]. Moreover, clinical supervision is knowingly effective in nursing practice [50]. Qin et al. [23] mentioned that effective clinical supervision enhances nursing

students' ethical sensitivity and decision-making. However, ineffective clinical supervision can hinder students' capacity to identify and resolve ethical sensitivity [51].

The third essential study result shows that nursing students rated their compassion competence highly. The nursing students' high compassion competence is shaped by integrating theory and its application to practice in the clinical environment and the exemplary behavior of nurses serving as role models [22]. As Younas and Maddigan [52] mentioned, establishing a policy framework in nursing education that institutes emotional learning as a top priority, encourages reflective practices, and incorporates empathetic care in clinical exposure cultivates the compassion competence of nursing students. These practices can develop compassionate nursing care that combines empathy, personalized care, and therapeutic communication [20].

The study further shows that female students exhibited better ethical sensitivity than men. This study result supports the finding that women are more inclined to demonstrate empathy and compassion than males [53]. This notion may be why the gender of student nurses influences their compassion competence. Female nursing students' caregiving and nurturing careers make them more compassionate [54].

The last essential result of the study shows that the “process subscale of clinical supervision” significantly influences student compassion competence. The “process subscale of clinical supervision” promotes self-reflection, feedback, and direction and helps students enhance clinical skills [29]. This involves working with patients and resolving ethical issues. The perceived effective, compassionate care and supportive supervisors should help student nurses develop self-awareness, emotional regulation, and empathy, which are essential to compassion competency [46].

The compassion competency of nursing students is greatly affected by the “impact subscale of clinical supervision.” The term “impact subscale of clinical supervision” refers to how students' experiences with clinical supervision affect their personal, professional, and educational development [29]. The clinical supervision of the instructors to the student nurses makes them feel more driven, valued, and ready to provide caring and considerate healthcare provider [55]. Thus, students can learn to give compassionate care under supervision in a safe, trustworthy setting that promotes open communication.

The connection between compassion competence among nursing students and the ethical sensitivity of the healthcare profession is inherent. Compassion

competency often requires making moral judgments that improve the dignity and well-being of patients [56]. Nursing students who comprehend ethical considerations are more likely to demonstrate compassion [57]. They possess sufficient sensitivity to discern the ethical importance of prioritizing patients. One can improve their ethical awareness by engaging in self-education, practicing reflection, and being exposed to ethical dilemmas in therapeutic environments [43]. Furthermore, communication proficiency and understanding patient issues are crucial aspects of compassion competence, as these elements should be prioritized [58].

Based on the study's result, it is recommended that clinical supervision should foster the development of nursing students' compassion and ethically sensitive care for their patients. Students engaged in nursing education should receive guidance in compassion competence and ethics through clinical supervision to enhance the quality of nursing they provide to their patients. Thus, the findings of this study imply that nurse education programs should prioritize implementing comprehensive clinical supervision frameworks. These frameworks include direct guidance, frequent constructive feedback, structured regular meetings, and a focus on personal, professional, and educational development. Adopting this framework will help improve the clinical practice of nursing students in providing ethical and compassionate care.

The curriculum development team should ensure that the knowledge and skills necessary for autonomy, the ability to make sound ethical decisions, and the capacity to manage complex ethical dilemmas are integrated into the course learning outcomes. Furthermore, nursing programs should incorporate ethics education and training to emphasize individual care strongly. Moreover, implementing gender-inclusive nursing education and support can develop heightened compassion and equity throughout the healthcare system.

4.1. Limitations of the study

The limitations of this research are noted in these components. The study's convenience sample consisted of nursing students from a single university. To improve representation, it is necessary to carry out further research in various regions and universities with varied populations. The data were obtained via self-report measures, which may have led to a social desirability bias and compromised the accuracy of the situation's portrayal. Nevertheless, the influence of this bias was probably reduced since the respondent remained anonymous. The

cross-sectional design made it difficult to determine the causal links between compassion competency, ethical sensitivity, and clinical supervision in nursing students. Conducting a longitudinal study to determine the causal links between clinical supervision, ethical sensitivity, and compassion competence among nursing students is essential. Also, a qualitative study should be conducted to explore the practice experiences of nursing students on clinical supervision, ethical sensitivity, and compassion competence.

5. Conclusion

The study shows that student nurses rated clinical supervision, ethical sensitivity, and compassion competence high. The study found that age, gender, and academic year level significantly influence students' assessments of their clinical supervision, ethical sensitivity, and compassion competency. In contrast, those in their higher academic years had a more positive perception of clinical supervision, while men showed a lower predisposition towards moral and compassionate attitudes. The highest average scores on the "Purpose" and "Process" subscales signify complete comprehension of the objectives and procedures of clinical supervision among student nurses. The lowest average score on the "Impact" subscale indicates the need for enhanced supervision sessions to attain the desired outcomes.

Ethics information

The Shaqra University Research Committee (ERC-20210041, March 6, 2022) ensured the ethical conduct of the study.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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